PRAKTIKUM BASIS DATA

MODUL 9

SCRIPTING PROCEDURE AND FUNCTION

Disusun Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

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Diberikan Query sebagai berikut:

QUERY

```
CREATE TABLE hremployees (
    employee_id    NUMBER(6)      NOT NULL,
    first_name     VARCHAR2(20),
    last_name      VARCHAR2(25)  NOT NULL,
    email          VARCHAR2(25)  NOT NULL,
    phone_number   VARCHAR2(20),
    hire_date      DATE          NOT NULL,
    job_id         VARCHAR2(10)  NOT NULL,
    salary         NUMBER(8,2),
    commission_pct NUMBER(2,2),
    manager_id     NUMBER(6),
    department_id  NUMBER(4),
    CONSTRAINT emp_emp_id_pk PRIMARY KEY (employee_id)
);

-- INSERT data untuk department_id = 90 (Executive — dipakai soal no.4)
INSERT INTO hremployees VALUES (100, 'Steven',  'King',     'SKING',    '515.123.4567', TO_DATE('17-JUN-2003','DD-MON-YYYY'), 'AD_PRES', 24000, NULL, NULL, 90);
INSERT INTO hremployees VALUES (101, 'Neena',   'Kochhar',  'NKOCHHAR', '515.123.4568', TO_DATE('21-SEP-2005','DD-MON-YYYY'), 'AD_VP',   17000, NULL, 100, 90);
INSERT INTO hremployees VALUES (102, 'Lex',     'De Haan',  'LDEHAAN',  '515.123.4569', TO_DATE('13-JAN-2001','DD-MON-YYYY'), 'AD_VP',   17000, NULL, 100, 90);

-- INSERT tambahan department lain (agar SELECT * tidak kosong)
INSERT INTO hremployees VALUES (103, 'Alexander', 'Hunold',  'AHUNOLD',  '590.423.4567', TO_DATE('03-JAN-2006','DD-MON-YYYY'), 'IT_PROG',  9000,  NULL, 102, 60);
INSERT INTO hremployees VALUES (104, 'Bruce',     'Ernst',   'BERNST',   '590.423.4568', TO_DATE('21-MAY-2007','DD-MON-YYYY'), 'IT_PROG',  6000,  NULL, 103, 60);
INSERT INTO hremployees VALUES (107, 'Diana',     'Lorentz', 'DLORENTZ', '590.423.5567', TO_DATE('07-FEB-2007','DD-MON-YYYY'), 'IT_PROG',  4200,  NULL, 103, 60);
INSERT INTO hremployees VALUES (124, 'Kevin',     'Mourgos', 'KMOURGOS', '650.123.5234', TO_DATE('16-NOV-2007','DD-MON-YYYY'), 'ST_MAN',   5800,  NULL, 100, 50);
INSERT INTO hremployees VALUES (141, 'Trenna',    'Rajs',    'TRAJS',    '650.121.8009', TO_DATE('17-OCT-2003','DD-MON-YYYY'), 'ST_CLERK', 3500,  NULL, 124, 50);
INSERT INTO hremployees VALUES (149, 'Eleni',     'Zlotkey', 'EZLOTKEY', '011.44.1344.429018', TO_DATE('29-JAN-2008','DD-MON-YYYY'), 'SA_MAN',   10500, .2, 100, 80);
INSERT INTO hremployees VALUES (174, 'Ellen',     'Abel',    'EABEL',    '011.44.1644.429267', TO_DATE('11-MAY-2004','DD-MON-YYYY'), 'SA_REP',   11000, .3, 149, 80);
```

OUTPUT

```
Table HREMPLOYEES created.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.

1 row inserted.
```

Analisis:

Kode ini membuat tabel hremployees sebagai sumber data utama.

1) employee_id sebagai primary key → tidak boleh duplikat dan tidak boleh null

2) Kolom seperti last_name, email, hire_date, job_id wajib diisi

3) Tipe data sudah disesuaikan:

a) VARCHAR2 untuk teks

b) NUMBER untuk angka

c) DATE untuk tanggal

Data yang diinsert mencakup beberapa department:

a. 90 (Executive)

b. 60 (IT)

c. 50 (Shipping)

d. 80 (Sales)

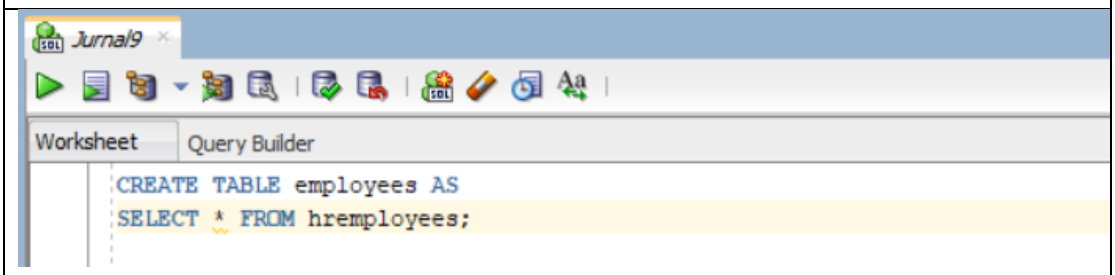
Artinya:

a. Data sudah cukup variatif untuk pengujian query

b. Ada relasi sederhana antar employee lewat manager_i

1. **CREATE TABLE EMPLOYEES AS SELECT * FROM HREMPLOYEES**

QUERY



OUTPUT

```
Table EMPLOYEES created.
```

Analisis:

Menduplikasi tabel hremployees ke employees.

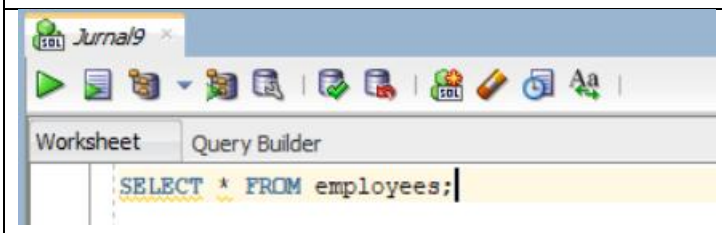
a. Struktur dan data ikut tersalin

b. Constraint seperti primary key **tidak ikut tersalin**

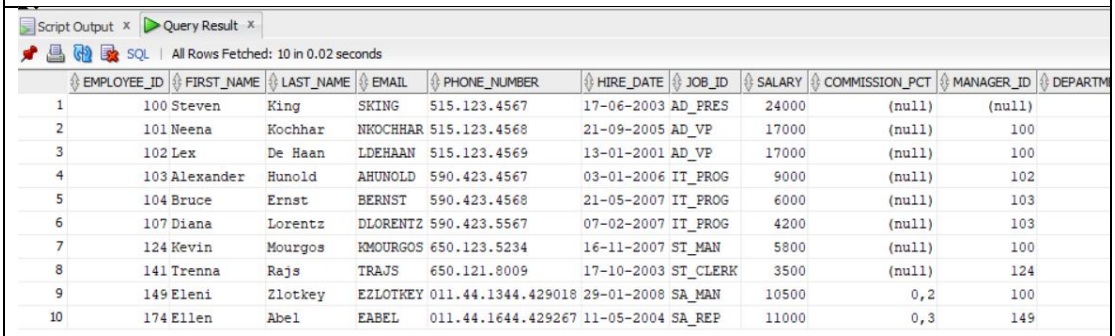
c. Tabel baru tidak terhubung lagi dengan tabel lama

2. SELECT * FROM EMPLOYEES

QUERY



OUTPUT



	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTM
1	100	Steven	King	SKING	515.123.4567	17-06-2003	AD_PRES	24000	(null)	(null)	
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-09-2005	AD_VP	17000	(null)	100	
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-01-2001	AD_VP	17000	(null)	100	
4	103	Alexander	Hunold	AHUNOLD	590.423.4567	03-01-2006	IT_PROG	9000	(null)	102	
5	104	Bruce	Ernst	BERNST	590.423.4568	21-05-2007	IT_PROG	6000	(null)	103	
6	107	Diana	Lorentz	DLORENTZ	590.423.5567	07-02-2007	IT_PROG	4200	(null)	103	
7	124	Kevin	Mourgos	KMOURGOS	650.123.5234	16-11-2007	ST_MAN	5800	(null)	100	
8	141	Trenna	Rajs	TRAJS	650.121.8009	17-10-2003	ST_CLERK	3500	(null)	124	
9	149	Eleni	Zlotkey	EZLOTKEY	011.44.1344.429018	29-01-2008	SA_MAN	10500	0,2	100	
10	174	Ellen	Abel	EABEL	011.44.1644.429267	11-05-2004	SA_REP	11000	0,3	149	

Analisis:

Menampilkan seluruh isi tabel employees.

a. Data sama dengan hremployees

b. Digunakan untuk verifikasi hasil cloning

c. Tidak ada perubahan data

3. MEMBUAT PROCEDURE INCREASE_SALARY

QUERY

```oracle_plsql
CREATE OR REPLACE PROCEDURE increase_salary (
    department_id_in IN employees.department_id%TYPE,
    increase_pct_in  IN NUMBER
)
IS
BEGIN
    FOR employee_rec IN (
        SELECT employee_id
        FROM employees
        WHERE department_id = department_id_in
    )
    LOOP
        UPDATE employees emp
        SET emp.salary = emp.salary + (emp.salary * increase_pct_in)
        WHERE emp.employee_id = employee_rec.employee_id;

        DBMS_OUTPUT.PUT_LINE('Updated ' || SQL%ROWCOUNT);
    END LOOP;
END;
/
```

OUTPUT

```
Procedure INCREASE_SALARY compiled
```

Analisis:

Procedure ini menaikkan gaji berdasarkan department.

a. Menggunakan loop per employee

b. Menghindari update massal langsung

c. increase_pct_in sebagai persentase kenaikan

d. Output menunjukkan jumlah row yang diupdate

4. MENJALANKAN PROCEDURE + ROLLBACK

QUERY

```
BEGIN
    increase_salary(90, .50);
    ROLLBACK;
END;
/
SELECT * FROM employees WHERE department_id = 90;
```

OUTPUT

```
Procedure INCREASE_SALARY compiled

PL/SQL procedure successfully completed.

EMPLOYEE_ID FIRST_NAME           LAST_NAME                 EMAIL                     PHONE_NUMBER         HIRE_DATE  JOB_ID        SALARY COMMISSION_PCT MANA
----------- -------------------- ------------------------- ------------------------- -------------------- ---------- ---------- ------ -------------- ----
        100 Steven               King                      SKING                     515.123.4567         17-06-2003 AD_PRES        24000
        101 Neena                Kochhar                   NKOCHHAR                  515.123.4568         21-09-2005 AD_VP          17000
        102 Lex                  De Haan                   LDEHAAN                   515.123.4569         13-01-2001 AD_VP          17000
```

Analisis:

a. Gaji sempat dinaikkan 50%

b. Tapi langsung dibatalkan oleh ROLLBACK

Hasil akhir:

a. Tidak ada perubahan di database

b. SELECT tetap menampilkan gaji lama

5. APA YANG TERJADI JIKA ROLLBACK DIHAPUS

QUERY

```
BEGIN
    increase_salary(90, .50);
END;
/
SELECT * FROM employees WHERE department_id = 90;
```

OUTPUT

```
PL/SQL procedure successfully completed.

EMPLOYEE_ID FIRST_NAME           LAST_NAME                EMAIL                    PHONE_NUMBER         HIRE_DATE  JOB_ID        SALARY COMMISSION_PCT MANA
----------- -------------------- ------------------------ ------------------------ -------------------- ---------- ---------- ---------- -------------- ----
        100 Steven               King                     SKING                    515.123.4567         17-06-2003 AD_PRES        81000
        101 Neena                Kochhar                  NKOCHHAR                 515.123.4568         21-09-2005 AD_VP          57375
        102 Lex                  De Haan                  LDEHAAN                  515.123.4569         13-01-2001 AD_VP          57375
```

Analisis:

Tanpa ROLLBACK:

a. Perubahan gaji benar-benar tersimpan
b. Salary employee department 90 naik 50%
c. Data berubah permanen

Kesimpulan:

a. ROLLBACK = batalin transaksi
b. Tanpa ROLLBACK = commit otomatis saat session selesai atau manual commit